Zayaan Bhanwadia

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Education

University of Toronto Sept. 2025 – June 2029

HBSc in Computer Science – Software Engineering Specialist, Statistics Minor (Co-op) | GPA: 3.7/4.0

- **Activities & Awards:** Dean's List, 3x Hackathon Winner, UTRA Web Developer, VP Technology @ CS Association
- **Coursework:** Software Design, Systems Programming, Data Structures & Algorithms, Computational Theory

Experience

AMD May 2027 – Aug. 2027

Incoming Software Engineer Intern (HIP Runtime) *Markham, ON*

Autodesk Jan. 2027 – Apr. 2027

Incoming Software Developer Intern (Backend/Platform) *Toronto, ON*

FreshBooks Sept. 2026 – Present

Software Developer Intern (Full Stack) *Toronto, ON*

- Shortened bank page load time by **86% (700ms to 100ms)** by moving a blocking API request to the background.
- Cut latency by **92%** over **20 benchmark calls** by caching invoices and paginated data in Redis with TTL and LRU.
- Built backend audit logging for **5** bank connection event types, giving users and Support a clear connection history.
- Unified Plaid and Yodlee bank connections across backend services and Ember for staged rollout to **100%** of users.
- Prototyped row-level locks on IDs and duplicates to replace **2 system-level locks** for concurrent bank imports.

University of Toronto Formula Racing (Driverless) July 2026 – Present

Autonomy Software Engineer (Mapping & Planning) *Toronto, ON*

- Developed mapping and path-planning software in C++ and ROS 2 for **Canada's first driverless FSAE car**.
- Reduced localization compute time by **30%** by replacing FastSLAM with EKF-SLAM while maintaining accuracy.
- Built ego-aligned track corridors from quintic splines using arc-length sampling across a configurable **15m** horizon.
- Integrated **8 sensors**, including IMU, wheel speed, and GPS, expanding simulation data collection and monitoring.

FlyRank AI July 2026 – Sept. 2026

Software Engineer Intern (Backend AI) *Toronto, ON*

- Prevented duplicate charges under **50+** simultaneous requests by deduplicating payment events in the database.
- Lowered spam submissions by **80%** by securing a public API with validation, rate limiting, and geolocation checks.

Projects

Embedded Key-Value Store | *C++, CMake, Multithreading, Linux*

- Built an embedded database with **16-way** sharded locking, boosting read throughput **5x (8M to 40M ops/sec)**.
- Implemented reader-writer locks so reads proceed during disk writes, preventing race conditions across **16 threads**.
- Built background compaction to purge stale and deleted keys, cutting disk usage by **45%** without pausing operations.
- Implemented crash recovery by persisting every update to a checksummed write-ahead log before applying changes.

Context Sync Extension | *TypeScript, VS Code Extension API*

- Developed a VS Code extension with **900+ downloads** that automates AI chat context sharing across environments.
- Raised recall from **67%** to **93%** on **8 synthetic cases** by combining lexical retrieval with weighted Dijkstra search.
- Compacted chat summaries by **30%+** by designing a Markdown schema to pack more context into fewer tokens.

Technical Skills

Languages: Python, Java, JavaScript, C++, C, TypeScript, SQL, Bash, HTML/CSS

Libraries/Frameworks: FastAPI, Django, Express.js, Node.js, React.js, Ember.js, SQLAlchemy, ROS 2

Developer Tools: Linux, Docker, Git, GitHub, Claude Code, Codex, Jira, Confluence

Cloud/Databases: PostgreSQL, MySQL, Redis, AWS, Azure, GCP, Vercel, Supabase, MongoDB